SEVEN EQUIVALENT CHARACTERIZATIONS OF CLOSED IMMERSIONS

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Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Source reference. Original lemma 05VH: sources/stacks-chow/tags.txt, source lines 7551–7551. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01L4: sources/stacks-chow/tags.txt, source lines 2073–2073. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02LS: sources/stacks-chow/tags.txt, source lines 3419–3419. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03BB: sources/stacks-chow/tags.txt, source lines 4342–4342. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01W5: sources/stacks-chow/tags.txt, source lines 2470–2470. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01L7: sources/stacks-chow/tags.txt, source lines 2076–2076. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. The following are equivalent:*

(1) *f is a closed immersion,*
(2) *f is a proper monomorphism,*
(3) *f is proper, unramified, and universally injective,*
(4) *f is universally closed, unramified, and a monomorphism,*
(5) *f is universally closed, unramified, and universally injective,*
(6) *f is universally closed, locally of finite type, and a monomorphism,*
(7) *f is universally closed, universally injective, locally of finite type, and formally unramified.*

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Proof. The equivalence of (4) – (7) follows immediately from Lemma 05VH.

Let $f : X \to S$ satisfy (6). Then f is separated, see Schemes, Lemma 01L4 and has finite fibres. Hence More on Morphisms, Lemma 02LS shows f is finite. Then Morphisms, Lemma 03BB implies f is a closed immersion, i.e., (1) holds.

Note that (1) $\Rightarrow$ (2) because a closed immersion is proper and a monomorphism (Morphisms, Lemma 01W5 and Schemes, Lemma 01L7). By Lemma 05VH we see that (2) implies (3). It is clear that (3) implies (5). $\square$